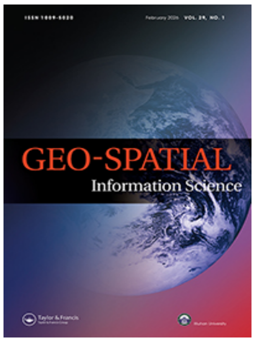

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Advancing unmanned roadheader-based tunneling: a framework for challenging underground environments

Changqing Yan[a], Gang Zhao[b,c], Shengzhe Hou[d] and Xinming Lu[d]

[a]College of Intelligent Equipment, Shandong University of Science and Technology, Tai'an, China; [b]State Key Laboratory of Soil Erosion and Dryland Farming on the Loess Plateau, Northwest A&F University, Yangling, China; [c]BASF Digital Farming GmbH, Cologne, Germany; [d]College of Computer Science and Engineering, Shandong University of Science and Technology, Qingdao, China

ABSTRACT

Unmanned roadheader-based tunneling in complex underground environments represents a significant advancement in mining technology, offering the potential to enhance safety and efficiency, but it poses considerable challenges in terms of precise control, navigation, and real-time decision-making. In response to these challenges, we have developed a comprehensive framework that integrates software, communication protocols, hardware, and newly developed algorithms. The proposed system architecture employs a laser-based total station for high-precision surveying, coupled with a programmable logic controller (PLC) integrated with multiple sensors to control the movement of the roadheader. To resolve key issues such as command issuance, robust positioning, autonomous navigation, real-time visualization of the working face, and trajectory planning for both the roadheader and cutter head, specialized algorithms have been developed. These components are integrated into a client/server software system, which includes a user-friendly interface that ensures efficient and safe tunneling operations. The client, deployed remotely, generates, and delivers operational instructions while providing real-time visualization of the working face, whereas the server, stationed on-site, executes these instructions, and ensures precise control and positioning of the roadheader. Comprehensive experimentation and system deployment have demonstrated the framework's effectiveness, successfully resolving key issues such as accurate positioning, unmanned navigation, cutting control, and multiple-trajectory planning. This system not only facilitates successful unmanned tunneling but also offers substantial benefits to the mining industry in China, with potential applications in other unmanned tunneling machines and civil construction projects.

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Roadheader; unmanned tunneling; underground positioning; trajectory planning; real-time visualization; total station; multisensor; client/server

1. Introduction

Tunnels are commonly constructed for transportation purposes in mountainous areas and for underground coal mining operations (Zhu, Yan, and Liang 2019). The tunnel working face is the most dangerous and labor-intensive areas, where accidents and fatalities occur at a high frequency (Deshmukh et al. 2020; Wang et al. 2020; Zhu et al. 2022). The working conditions are extremely harsh, especially in the very deep underground, due to factors such as high-density dust, intensive vibration, low illumination, complicated backgrounds, high humidity, and complex rock mass and geological conditions (Stille and Palmström 2003). Unmanned tunneling can significantly improve safety by keeping workers away from hazardous and extreme environments (Zhang, Zhu, et al. 2023). Therefore, it is crucial to urgently implement unmanned operations to replace manual labor in such environments, which can not only protect the lives and health of miners but also enhance work efficiency. This need is particularly urgent in China, where tunneling for mining often extends to depths exceeding 1000 m and involves extremely complex underground conditions. The key to unmanned tunneling lies in the essential machinery used for tunnel excavation. The selection of the essential tunneling machinery could be different under different tunneling conditions (Ozfirat et al. 2015). Boom-type roadheaders are the primary machines used for excavating roadways, sewers, and mining projects (Deshmukh et al. 2020; Fu et al. 2018). In addition to the factors including power of the machine, weight of the machine, ratio of power to weight, drilling rate index, uniaxial compressive strength, and rock mass rating (Ozfirat et al. 2017), given China's heavy reliance on coal for energy, the automation, and robotization of roadheaders hold great significance for improving the tunneling efficiency and achieving unmanned tunneling, thereby meeting the country's huge coal requirements.

CONTACT Xinming Lu 13853803297@139.com

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However, realizing unmanned tunneling faces various challenges in the harsh underground environment. These challenges include magnetic disturbance, electrostatic interference, and the unavailability of GPS signals (Zhang, Hao, et al. 2023). Researchers have conducted studies to address these challenges and advance unmanned roadheader tunneling in recent decades (Deshmukh et al. 2020). These studies can be categorized into three main areas. The first category addresses fundamental challenges in the automation and robotization of roadheaders, particularly in positioning and pose measurement (Zhang, Hao, et al. 2023). To overcome these challenges, several methods have been proposed. One notable approach involves a real-time roadheader pose detection system that utilizes cross lasers and laser targets (Du et al. 2016). Another method combines a strapdown inertial navigation system with monocular vision for automatic position and pose measurement (Yang et al. 2018). Additionally, kinematic models and vision-based cutter head pose measurement methods have been developed (Tian, Wang, and Wu 2018; Yang et al. 2018). Despite the advancements represented by these methods, each face limitations when confronted with the challenges at hand. As of now, few widely applied positioning methods can achieve precise and robust positioning in harsh underground conditions in China, thus limiting their real-world applications. Recently, a promising multisensor-based method has been proposed to address these issues, while further testing is required for broader engineering applications (Yan, Zhao, and Lu 2019).

The second category focuses on the automatic control of the roadheader's cutter head, a subject which has been explored by several research studies. Cheluszka (2015) conducted a research to simulate the heading surface mining process using a computer with conical picks of cutter heads for performance enhancement of the cutter heads. Subsequently, Cheluszka, Sobota, and Gluszek (2019) developed an automatic cutter head control system, enabling higher efficiency and reduced energy consumption compared to manual methods. Additionally, the response of the roadheader's body pose to factors, such as the horizontal swing angle of the cutting arm, cutter head load, and dip angle of the coal seam during the cutting process has also been investigated, providing valuable insights for body pose adjustment and automatic cutting (Zong et al. 2018).

The last category focuses on the unmanned tunneling framework and software system. For example, an autonomous pilot system has been developed for navigation, self-localization, and path planning in tunnel boring machines (TBM) (Ferrein, Kallweit, and Lautermann 2012). In addition, guidance systems, 4D construction planning models, and real-time data processing solutions have been developed for TBM steering, automated data collection, and 3D visualization, thereby enhancing construction management (Liang and Lu 2010; Shen et al. 2014). However, these solutions have primarily been tested in straight line tunneling, with further validation required for curved tunnels. Moreover, a virtual laser target board system has also been proposed for real-time monitoring of the TBM working process (Mao et al. 2013). Yet, this system faces challenges related to laser dispersion issues and difficulties in the accurate installation of the employed prisms. In subsequent research, total station systems were introduced for accurate and real-time positioning, but their setup and relocation are complex, requiring survey experts, which limits their widespread use and diminishes positioning reliability (Mao et al. 2013; Shen et al. 2014). Additionally, a mathematical method based on kinematic analysis has been proposed to address the automatic control issue of shield tunneling machines (STM) by determining the target motion of the cylinders (Wang, Li, and Wu 2018). However, it is essential to acknowledge that the kinematics and structure of STM differ significantly from roadheaders. Despite the previously mentioned studies, a practical and well-established unmanned tunneling solution tailored specifically for roadheaders is currently lacking and demands further development in China.

The primary objective of this manuscript is to formulate an unmanned framework and integrated system that effectively addresses the challenges associated with roadheader-based tunneling. In this manuscript, based on our previous research on positioning methods and drawing inspiration from the relevant studies (Mao et al. 2013), a comprehensive unmanned solution for roadheaders is proposed and a corresponding software system is developed, overcoming critical issues such as visualization, information modeling, and automatic control. Experimental results demonstrate that our solution is well suited for engineering applications.

The manuscript is organized into five distinct sections. Section one reviews the relevant research on this topic. Section 2 introduces our solution for unmanned tunneling employing roadheaders. Section 3 shows the experiments and the results, while section 4 presents a discussion about the results. The last section concludes the work

2. Method

2.1. Empowering unmanned tunneling – a comprehensive framework

To achieve unmanned tunneling with roadheaders, it is imperative to integrate and connect essential components and instruments for surveying, control, execution, and computing. These elements come together to form a comprehensive and cohesive integrated framework, as depicted in Figure 1.

According to Figure 1, the architecture consists of multiple devices, with a switch connecting all the components. It includes two types of personal computers (PCs): an on-site PC near the working face and multiple remote PCs away from the working face or at the ground office. Connected to these PCs are various devices, such as remote controllers linked to the remote PCs, an inertial system, a compass, a total station (TS), a programmable logic controller (PLC) unit, and a roadheader linked with the on-site PC through the sensors.

The components are categorized into four primary parts: instruction or command issuers, data collection/command execution, position computing, and working status visualization (Figure 1). Part 1 can be a physical remote controller equipped on a ground PC or remote PC, or it can be a program simulating the functionality of a physical remote controller. Notably, multiple remote controllers can be deployed on corresponding PCs in different locations as needed to monitor the roadheader in various places. The physical remote controller enables manual operation from a distant location, while the simulated remote controller aims to enable automatic or unmanned operation without the support from the workers near the working surface. Both operation methods should coexist and can be switched as needed, enhancing the system's flexibility and convenience.

Part 2 comprises multiple sensors, including an inertial navigation system (INS), a compass, a TS, and a roadheader with multiple sensors for hydraulic cylinders and a PLC for working status measurement and control, as illustrated in Figure 2. The INS

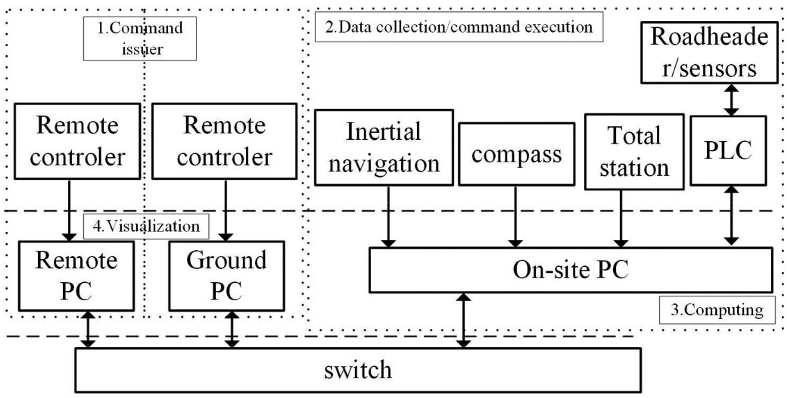

Figure 1. Architecture and key data flow of the intelligent tunneling framework.

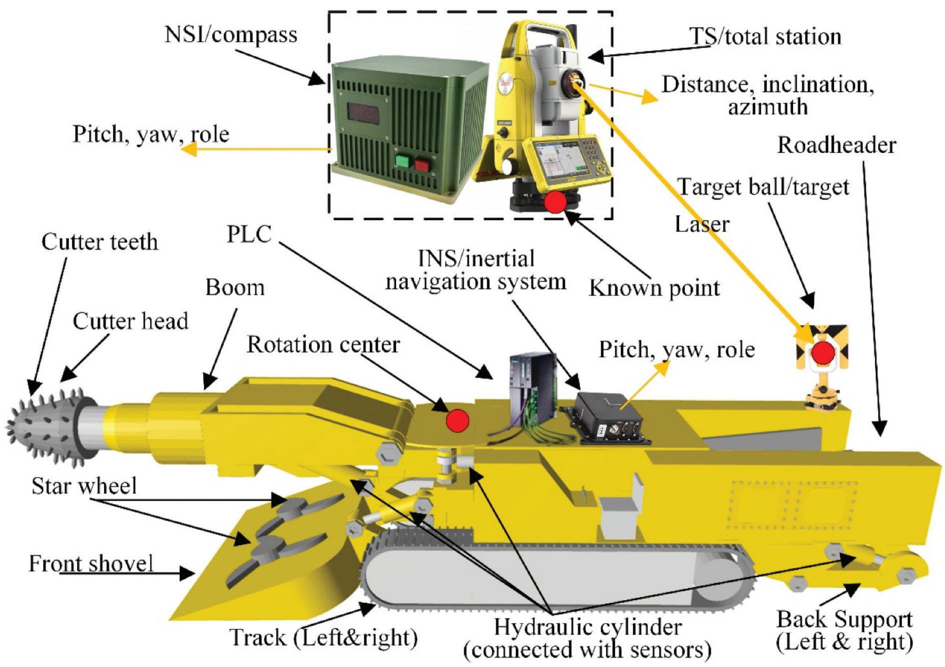

Figure 2. Roadheader and its key components, including total station (TS), compass, inertial navigation system (INS), and programmable logic controller (PLC) and their deployment.

measures real-time pose angles of the roadheader, and the compass provides pose angles for the TS (Figure 2), both of which serve similar functions. The former is more affordable and outputs data at longer intervals (one or two seconds), while the latter is more expensive but outputs data at shorter intervals (about 50 ms) to meet the requirement of measuring changing pose angles in real-time. The TS is used for position surveying and can track a moving target ball fixed at the rear of the roadheader (Figure 2). When combined, these devices collect the necessary data for absolute position computing. Due to the reasonable rigid-body assumption of the roadheader, once a single-point position is computed, any point on the roadheader can be determined by the relative position that could be directly measured with respect to this point. To position the cutter head, the momentary hydraulic cylinder lengths must be measured by connected sensors and recorded by PLC, and then used for position computing.

Based on Parts 1 and 2, Part 3 primarily involves the on-site PC responsible for positioning computing using the information collected by the multisensor system, while Part 4 focuses on working status and environment monitoring visualization. With Part 4, the tunneling status and the roadheader status can be clearly observed by users.

2.2. Streamlining unmanned tunneling: a comprehensive workflow

Building upon the above introduced framework, this section presents a meticulously designed workflow to achieve unmanned tunneling (Figure 2). In this context, RPC and SMR denote remote personal computers and on-site personal computers, respectively. RPC serves as remote controllers and monitoring systems, while SMR serves as a system for unmanned roadheader driving and positioning.

As depicted in Figure 3, the workflow commences with the input of sectional information, including the tunnel's shape, and the tunnel axis information, into the RPC. Subsequently, trajectory planning is executed to guide the movement of the roadheader and its cutter head, to build the tunnel. The obtained paths serve as a roadmap for steering the roadheader, while the cutter head follows a designated path after the roadheader reaches the target point on the trajectory. The RPC continuously checks and compares the current position and posture of the roadheader with the target point in the planned path to make informed decisions for subsequent movements. Once a decision is made, the instruction is transmitted to the SMR, which is then executed to control the roadheader accordingly. Concurrently, data from the sensors is collected by the SMR to compute the precise position of the roadheader and cutter head. The SMR then provides the RPC with positioning information for visualization.

The workflow revolves around four key functions: instruction generation and delivery, roadheader positioning and navigation, trajectory planning, and working face status visualization. These functions synergistically contribute to the successful realization of unmanned tunneling, minimizing human intervention, and ensuring seamless and efficient tunnel construction.

2.2.1. Mechanism for instruction generation and delivery

The generation and delivery of instructions play a pivotal role in driving the roadheader-based unmanned tunneling system. These instructions are generated originally by a physical remote controller. However, this method requires the deployment of a device and is limited by the endurance of the hardware itself, which adds up the complexity and the cost. To replace the physical remote controller with

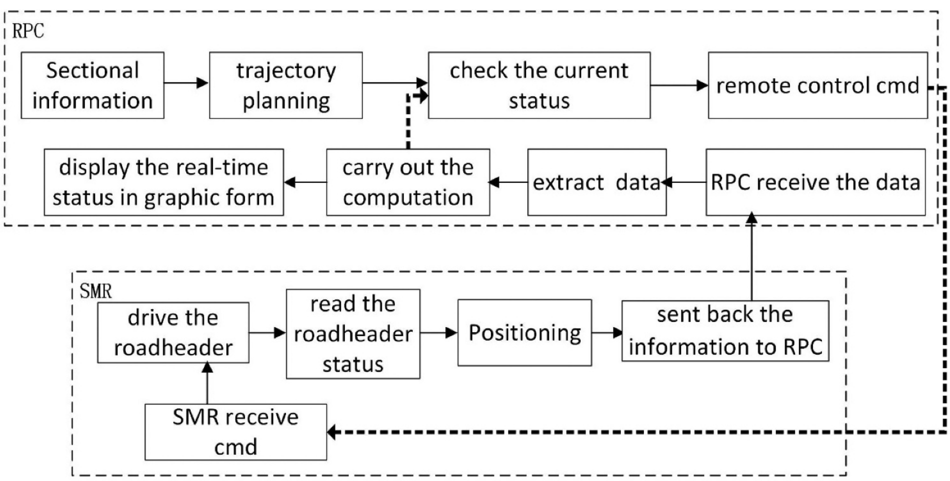

Figure 3. Workflow for the unmanned tunneling process.

a program, the formats of the instructions are analyzed and redesigned. The instruction follows a modbus protocol format (Tamboli et al. 2015). The data part of the instruction was redesigned, using 12 bytes for controlling the roadheader. Among these 12 bytes, the first 4 bytes are allocated for common functions, such as stop and halt, etc., along with other operations that facilitate the system's functionality. The second 4 bytes are for the components control, and the last 4 bytes for the roadheader motion and the cutter motion control. Since these 8 bytes are critical for the working status monitoring and the roadheader control, they are listed in Table 1. Based on the designed instructions, a program is developed to simulate the physical remote controller, enhancing the operative flexibility. Further, a fully automatic instruction generation method is also developed for automatic tunneling, tailored to the specific demands of tunneling process.

All the three methods follow a similar process for instruction delivery. Instructions are first generated based on tunneling requirements – either by a physical remote controller, a program-based remote controller, or an automatic mode. These instructions are then transmitted from the RPC to the SMR for roadheader control.

Although the physical remote controller mode and the program-simulated remote controller differ, both are classified as manual operation methods. To accommodate user preferences and the complexity of underground conditions, both manual operation methods and an automatic operation mode are provided in our system. These modes allow for seamless switching between them, ensuring flexibility and ease of use.

2.2.2. Roadheader positioning and navigation

2.2.2.1. Positioning method. To achieve precise positioning, a robust approach is developed by integrating data from the compass device, inertial navigation device, and TS device. As previously discussed, the roadheader can be modeled as a rigid body. Once the coordinates and orientation of one point are established, and the relative position of an unknown point with respect to the known point is determined, the location of the unknown point can be accurately calculated. According to this positioning principle, the positioning in tunneling can be accomplished. The whole positioning process is illustrated in Figure 4.

First, a point is surveyed under the earth coordinate system and designated as the known point, serving as the TS origin. Then, the pose angles of this point are measured. Using the known point and the pose angles,

Table 1. Instruction design for controlling the roadheader and its components.

Address	Function	Value (hex)	bit
0002	Dust fan start	0001	bit0
	Dust fan stop	0002	bit1
	Oil pump motor start	0004	bit2
	Oil pump motor stop	0008	bit3
	Spray pump start	0010	bit4
	Spray pump stop	0020	bit5
	Left rear support up	0100	bit8
	Left rear support down	0200	bit9
	Shovel plate up	0400	bit10
	Shovel plate down	0800	bit11
	Right rear support up	1000	bit12
	Right rear support down	2000	bit13
	Alarm bell start	4000	bit14
	System reset	8000	bit15
0003	Cutting height motor start	0001	bit0
	Cutting depth motor start	0002	bit1
	Secondary conveyor motor start	0004	bit2
	Secondary conveyor motor stop	0008	bit3
	Left star wheel forward start	0010	bit4
	Left star wheel forward stop	0020	bit5
	Left star wheel reverse start	0040	bit6
	Left star wheel reverse stop	0080	bit7
	Right star wheel forward start	0100	bit8
	Right star wheel forward stop	0200	bit9
	Right star wheel reverse start	0400	bit10
	Right star wheel reverse stop	0800	bit11
	Primary conveyor forward start	1000	bit12
	Primary conveyor forward stop	2000	bit13
	Primary conveyor reverse start	4000	bit14
	Primary conveyor reverse stop	8000	bit15
0004	Roadheader right turn	0x0080-0x00ff	bit0-bit7
	Roadheader left turn	0x007e-0x0000	bit0-bit7
	Roadheader forward	0x8000-0xff00	bit8-bit15
	Roadheader backward	0x7e00-0x0000	bit8-bit15
0005	Boom right turn	0x0080-0x00ff	bit0-bit7
	Boom left turn	0x007e-0x0000	bit0-bit7
	Boom up	0x8000-0xff00	bit8-bit15
	Boom down	0x7e00-0x0000	bit8-bit15

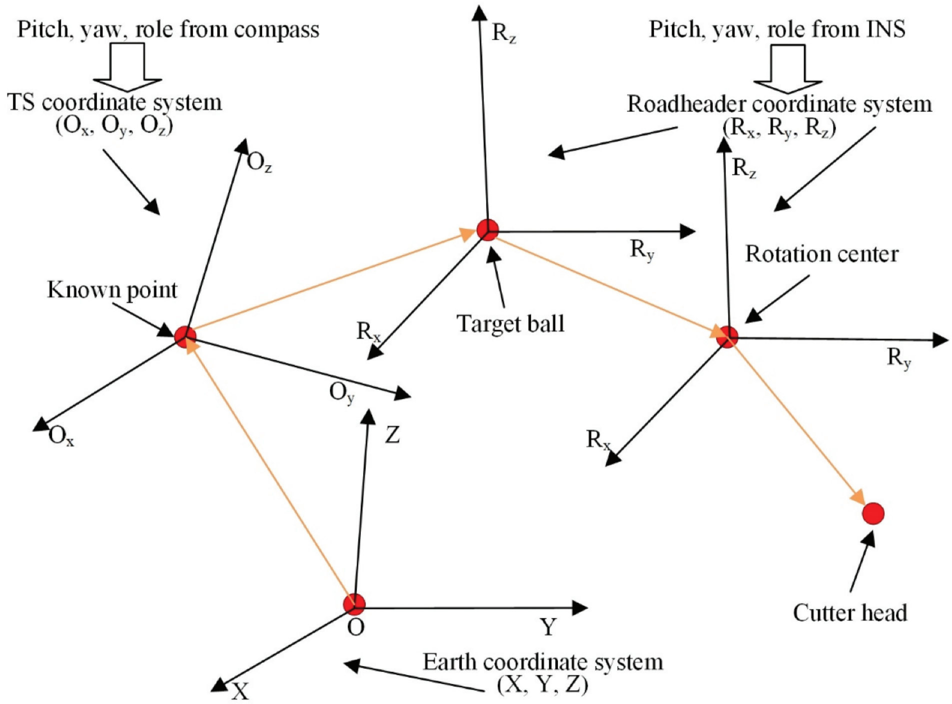

Figure 4. Workflow for the positioning process through a series of coordination transformation.

the position of the target ball can be determined based on the relative position acquired from TS. Then, the rotation center of the roadheader can be determined in a similar way. Using the known rotation center and the relative position with respect to it, the cutter head can be positioned.

The roadheader positioning process can be further described as follows.

First, the coordinate system (o_x, o_y, o_z) of TS should be calculated according to the following expressions (1), (2), and (3).

$$o_x = \begin{cases} x = \cos(yaw)\cos(pitch) \\ y = \sin(yaw)\cos(pitch) \\ z = \sin(pitch) \end{cases} \quad (1)$$

$$o_y = \begin{cases} x = -\cos(yaw)\sin(pitch)\sin(roll) - \sin(yaw)\cos(roll) \\ y = -\sin(yaw)\sin(pitch)\sin(roll) + \cos(yaw)\cos(roll) \\ z = \cos(pitch)\sin(roll) \end{cases} \quad (2)$$

$$o_z = \begin{cases} x = -\cos(yaw)\sin(pitch)\cos(roll) + \sin(yaw)\sin(roll) \\ y = -\sin(yaw)\sin(pitch)\cos(roll) - \cos(yaw)\sin(roll) \\ z = \cos(pitch)\sin(roll) \end{cases} \quad (3)$$

where (o_x, o_y, o_z) represent coordinate axis vectorso_x, o_y, o_z, and yaw, pitch, and roll are pose angles from the compass.

Then, the relative position x_{ob}, y_{ob}, z_{ob} of the known point with respect to the TS position can be calculated.

$$\begin{cases} x_{ob} = L \times \cos\alpha_q \times \sin\alpha_f \\ y_{ob} = L \times \cos\alpha_q \times \cos\alpha_f \\ z_{ob} = L \times \sin\alpha_q \end{cases} \quad (4)$$

where the coordinates x_{ob}, y_{ob}, z_{ob} represent the coordinates of the target ball under the TS coordinate system, L for the measured distance, α_f for Azimuth and α_q for Inclination, provided by the TS.

Using the computed o_x, o_y, o_z,x_{ob}, y_{ob}, z_{ob} and the known point (x_o, y_o, z_o) as origin, the fixed point with the target ball can be positioned according to (5).

$$(x_{eb}, y_{eb}, z_{eb}) = (x_{ob} \times o_x + y_{ob} \times o_y + z_{ob} \times o_z) + (x_o, y_o, z_o) \quad (5)$$

where (x_{eb}, y_{eb}, z_{eb}) represent the coordinates of the unknown position.

In a similar process, as the most critical point for cutter head positioning, the rotation center, illustrated in Figure 2, is determined.

As for the positioning of the cutter head, its relative position regarding the rotation center under the roadheader coordinate system is determined using data from the sensors, specifically, the hydraulic cylinder length. The roadheader coordinate system is determined by the precise pose angles measured by a high-accuracy INS integrated into the roadheader. Although the positioning process for the cutter head is similar to that of the roadheader, it is more complex due to the computation of the relative position using hydraulic cylinder length. For more details of this cutter head positioning process, readers can turn to the related work (Cheluszka 2015; Yan, Zhao, and Lu 2019).

2.2.2.2. Autonomous navigation and automatic cutting. Achieving unmanned tunneling requires autonomous navigation and cutting control of the roadheader based on the progress of tunneling tasks. To achieve this, the roadheader's forward direction is first adjusted to align with the next target point on the tunnel central axis. Subsequently, the roadheader is driven forward until its center reaches the target point, followed by the movement of the cutter head. This autonomous navigation and automatic process are illustrated in Figure 5.

According to Figure 5, this process can be detailed as follows: First, the angle is calculated between the current forward direction and the vector connecting the roadheader's rotation center to the target point, and then the distance between the current position and target position is calculated. With the calculated angle and distance, the specific operations to follow are planned. Subsequently, instructions are generated and issued to rotate the roadheader to align the forward direction with the target point. After that, instructions are issued to drive the roadheader forward or backward until the target point is reached. Upon reaching the target point, necessary operations are made to prepare the roadheader for cutting. Finally, the cutter head is controlled to carry out cutting until the task is accomplished.

The control of the roadheader's cutter head involves writing the expected hydraulic cylinder length to the PLC to either extend or retract the cylinder, thereby guiding the movement of the cutter head. As mentioned earlier, the position of the cutter head is determined by the relative position of the rotation center on the roadheader, which is calculated based on measured cylinder lengths. Conversely, to control the cutter head's movement, the positional information is converted back to the corresponding hydraulic cylinder lengths, which is then used to guide the boom movement (Figure 2). The interested readers can turn to the related work for more details (Yan, Zhao, and Lu 2019).

2.2.2.3. Handling target ball loss. In a challenging tunneling environment, characterized with dense dust and substantial vibrations resulting from the cutting operation, TS, the critical surveying device, may occasionally lose target tracking, potentially leading to hazardous cutting. To mitigate this issue and ensure reliable positioning, two corrective measures have been implemented.

The first one is an optimized re-searching method for the target ball. To implement this measure, the target ball status should be checked continuously during working time. Once the target ball is lost, the system should halt the roadheader's motion first and timely initiate a re-search for the target ball. If the target ball remains elusive after the initial search, an incremental angle range scanning is conducted. Once the target ball is found, the positioning process resumes, and the operation continues as planned. Otherwise, the roadheader comes to a complete stop.

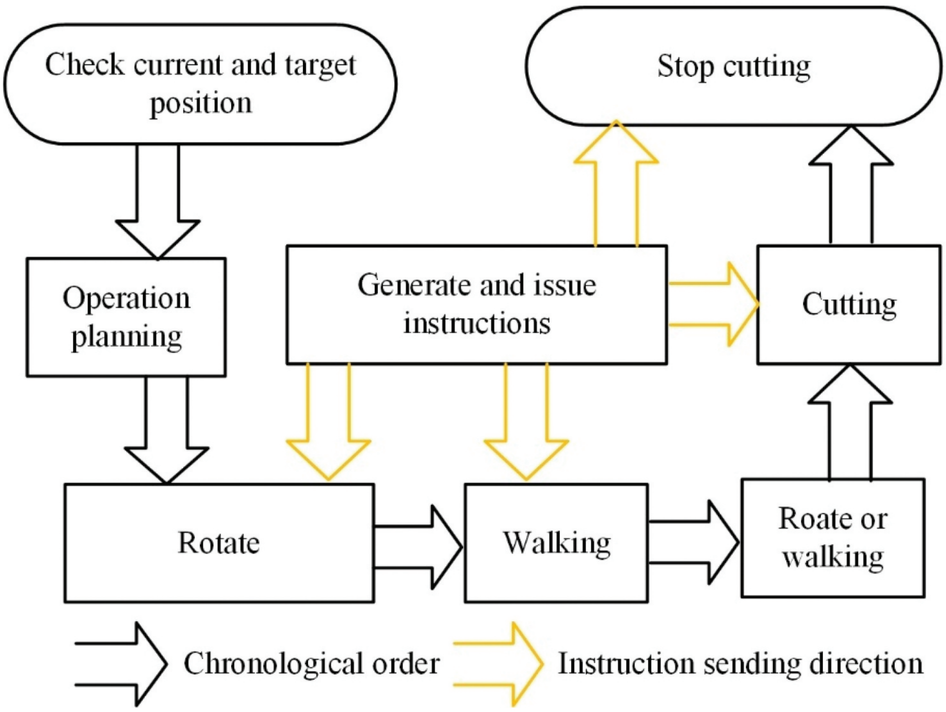

Figure 5. Workflow for autonomous navigation and cutting control.

To efficiently locate the lost target ball, the possible position is predicted using historical data on its previous location and movement speed, thereby accelerating the search process. This prediction narrows down the search area and accelerates the search process.

The second measure is to eliminate the dense dust that might prevent the target ball from being tracked. To meet this requirement, a blower is employed to clear the dust. This prevents the target ball from being obscured, ensuring continuous tracking and positioning accuracy to a great extent.

2.2.3. Real-time visualization of the working face status

Accurate visualization of the working face status is paramount to faithfully simulate the dynamic tunneling process. Such a visualization involves precise modeling of key components, including the real-time cutting status model of the cutter head in the working face, the roadheader model and the tunnel model, and a well-designed visualization solution.

2.2.3.1. Cutting status modeling of the cutter head.

An intuitive and real-time visual representation of the cutting status of the cutter head is crucial for ensuring high accuracy in tunneling projects. By providing a clear and accurate depiction of the current operational state, such a system enables operators to make timely adjustments and informed decisions, thereby enhancing the overall accuracy of the cutting process. However, the traditional reliance on drivers' direct onsite observation of the cutting status, even with a strong light, leads to low accuracy and inconsistency. To address this issue, a real-time cutting status model is developed, incorporating real-world parameters. The process of this real-time model construction is described as follows.

First, the position of each cutter tooth tip is determined through two steps. The first step involves dividing the cutter head into three equal thickness sections using a plane perpendicular to the central axis, resulting in three circular sections and a cutter head tip point. In the second step, the boundary of each section is discretized into 12 evenly distributed vertices to accurately approximate the boundary, yielding a total of 37 tooth tip positions.

Second, the cutter teeth which have penetrated the working face are identified through the following steps. First, the normal vector at each vertex is calculated. Then, the angle between the normal vector and the forward direction of the tunnel central axis is calculated. Vertices with an angle of 90 degrees or less are retained, as they correspond to cutter teeth directly penetrating the working face.

Third, the penetration depth of the cutter teeth into the working face is determined by calculating the distance that each vertex extends into the reference plane of the working face.

Finally, the cutting status is visualized by assigning different colors to varying depths. Specifically, the depth at each position on the cutting status is mapped using a color discretization scheme, with colors ranging from 0 to 255. This color arrangement provides a clear representation of the cutting depth, with 0 representing the maximum depth and 255 the minimum. Areas where the cutter head exceeds this range are represented in blue.

Moreover, to enhance user convenience, we provide multiple cutter head presentation modes, including an ellipsoid-shaped mode, a convex that surrounds the actual cutter head, and the previously mentioned color-coded depth representation for the cutter teeth. These options are designed to accommodate various preferences and requirements.

2.2.3.2. Roadheader, working face, and tunnel modeling.

Given the complexity of roadheaders, nonessential details that do not impact cutting precision should be simplified. In contrast, critical geometric parameters that directly influence cutting accuracy should be meticulously calibrated to align with the actual roadheader. As previously mentioned, the roadheader is treated as a rigid body, with no consideration for shape deformation. In line with this principle, key points that represent the shape or essential operational functions are retained to create a simplified yet accurate roadheader model. Specifically, the main body is represented as a cube, while the boom is modeled using regular geometry defined by the centerline and other boundaries. Additional key points, such as the target ball, rotation center, and various joint points, are treated as operational reference points.

As for the tunnel and the working face, our tunnel modeling approach incorporates critical parameters that define a tunnel's geometrical characteristics. These parameters include the width and height (or multiheight for ladder-shaped tunnels), as well as the radius for arc-shaped tunnels, which together describe the cross-sectional profile of the working face. Furthermore, the central axis parameter is employed to define the tunnel's orientation or direction.

2.2.3.3. Real-time visualization.

To convey comprehensive and understandable information of the working status, both the exact status of the working face and the precise relationship between the roadheader and the tunnel should be provided. To display these kinds of information, four different view perspectives are designed, one for the working face, and the other three for illustrating the relationship between the roadheader and tunnel (Figure 6).

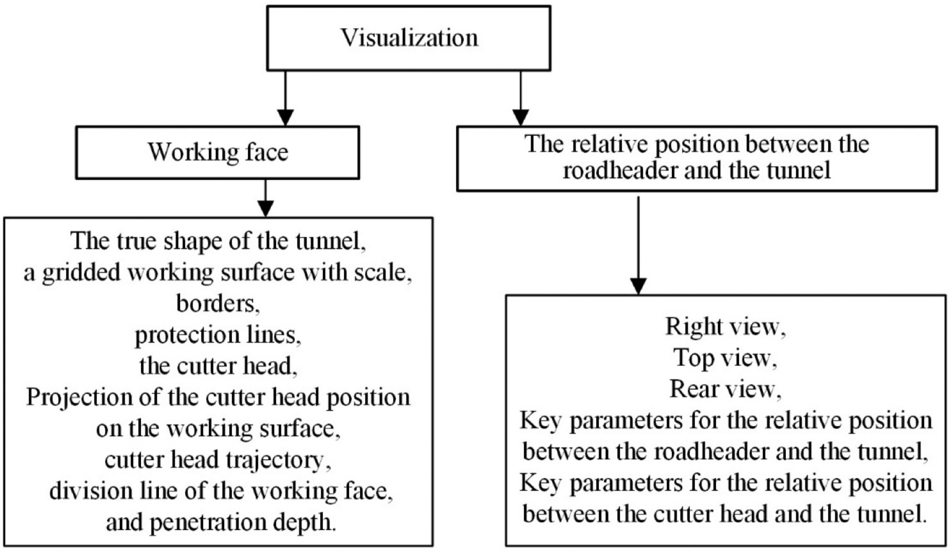

Figure 6. Components of visualization and elements to be visualized.

The view displaying the working face status serves as the core and primary view, providing users with information about the position of the cutter head on the working face and the progress made on the current working face. To accurately reflect the working status, it is essential to display the tunnel section boundaries, the trajectory covered by the cutter head, and the depth of penetration of the cutter head into the working face, as derived from the real-time cutter status model. Additionally, to further assist users, several other reference lines should be included, such as some self-defined lines demarcate a smaller area in the working surface for guiding the cutting process, a split line dividing the working face into two distinct cutting sections, and an alert line indicating when the cutter head approaches the border area, helping to prevent cutting beyond the designated boundaries.

The three views depicting the relationship between the roadheader and the tunnel, including right view, top view, and rear view, provide a comprehensive perspective on the relative position of the roadheader and cutter head with respect to the tunnel. In the right view, the depth of the cutter head's penetration into the working face is visible. Since tunnel cutting occurs in phases, with each phase involving cutting a few meters, followed by a pause for reinforcement and other tasks before the next cutting phase begins, it is important to indicate both the start and end points of the current phase. In our experimental area, a single cutting phase is further divided into two parts due to the cleaning work, which should be also illustrated clearly. A dotted black vertical line marks the beginning of the current cutting operation, while a dotted blue line indicates the end boundary. Dotted green lines are used to delineate the two parts of the cutting process within a single phase. Additionally, support rivets on the ceil of the tunnel near the working face are presented by short red-line segments.

The top view displays the relative position between the central axis of the roadheader and the tunnel, highlighting any deviation from the established track. This view aids users in inspecting the roadheader and making precise adjustments. Since the boom has its maximum width at the central joint, which could potentially obstruct movement within the tunnel, this point of maximum width is marked with a red line to facilitate early detection of any potential collision with the tunnel's boundary. Additionally, the lateral reach of the cutter head is shown, allowing users to assess whether the roadheader should be advanced further. The rear view illustrates the roadheader's roll relative to the plane perpendicular to the tunnel's central axis. Similar to the primary view, a red dotted line is used as an alert when the cutting position exceeds the border or any predefined distance threshold to it. To ensure that users can obtain precise information, the exact values of key variables should also be provided. These include roll, yaw, and pitch angles of the roadheader, the distance covered by the roadheader, horizontal distances from the cutter head center to various points, such as the front border, bottom, central axis, left, and right borders, as well as horizontal distances between the boom and the left and right borders.

These views provide a precise visualization of the roadheader's status and its exact relative position relationship.

2.2.4. Trajectory planning

To accomplish unmanned tunneling with roadheaders, flexible trajectory planning is indispensable for both the roadheader and the cutter head.

2.2.4.1. Roadheader trajectory planning. The roadheader must autonomously navigate along a predetermined trajectory to carry out the tunneling task effectively. This trajectory planning is based on the available tunnel information, including the central axis and sectional details, such as width, height, and other geometric parameters. The central axis is discretized into a series of trajectory points that the roadheader follows as it moves during the tunnel process (Figure 7).

The central axis typically comprises multiple interconnected line segments (Figure 7). To adapt these segments for the roadheader's cutting cycle, a discretization process is employed, including the following steps. First, using the start and end vertex of the first line segment of the central axis, the direction can be calculated. Then, with the start vertex, the direction, and the predefined discretization increment along the direction, the next point is determined in terms of Equation (6).

$$P_{discr} = P_{begin} + (P_{end} - P_{begin}) \cdot \Delta \tag{6}$$

where P_{discr} represents the obtained discretization point, P_{begin}, P_{end} and Δ represent the begin vertex, end vertex, and increment for discretization along the segment, respectively.

This operation is repeated until the end vertex is reached, completing the discretization of the line segment. Once all the line segments on the central axis have been discretized, the overall discretization of the central axis is complete, resulting in a final array of trajectory points.

While the roadheader moves along the trajectory points over a long distance, it is not always feasible to achieve strict alignment of the roadheader's center with the trajectory points due to potential accumulated positioning errors. As a result, a small margin of error is typically tolerated, and adjustments should be made when the error exceeds a predefined threshold.

2.2.4.2. Cutter head trajectory planning. Cutter head trajectory planning aims to design a path that ensures complete coverage of the entire cross-sectional surface, with sectional data serving as the foundation. Four planning methods are supported, offering flexibility in the cutter head's movement. One widely adopted approach is the S-shaped trajectory, which follows a series of connected line segments to cover the whole sectional area, similar to the central axis discretization. The second method is a spiral trajectory, which begins from the center and gradually expands outward in a spiral pattern. The third method involves manual trajectory planning through mouse clicks on the working face of the system interface, offering flexibility for customized path design. The last method captures the trajectory path by recording the path the cutter head follows under the guidance of workers. By incorporating these diverse trajectory planning techniques, we can effectively guide both the roadheader and cutter head throughout the tunneling process, ensuring high performance and accuracy in unmanned tunneling operations.

2.3. Interface design

To assist users in real-world tunneling, a friendly user interface providing comprehensive information is essential. To meet this need, we designed the interface shown in Figure 8, which can be divided into three main sections. The left section focuses on the operation setting, monitoring the work status of all major devices and displaying the current and press status information of the components of the roadheader (Figure 8A–C), allowing workers to easily assess the specific working conditions. The central section of the interface is divided into two main areas, the upper and lower portions (Figure 8D, E). The upper section provides additional information on operations and control, detailing processes such as the rotation and movement of the roadheader, selection of the cutter head trajectory, and the activation of fully automatic mode (Figure 8D). In this fully automatic mode, the roadheader autonomously adjusts its direction, navigates to a specified point, makes necessary positional

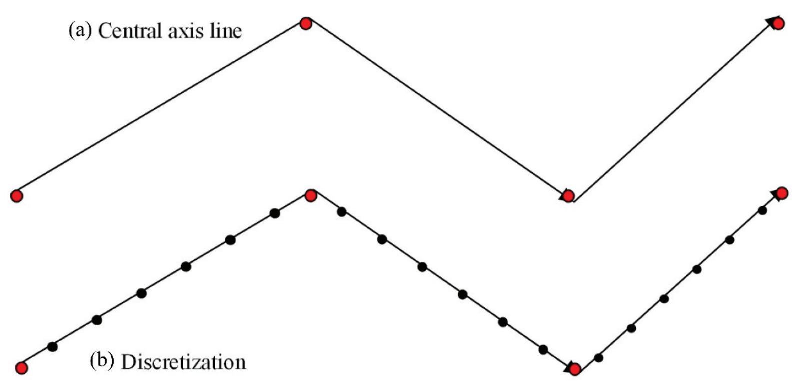

Figure 7. Discretization of the middle axis of a tunnel.

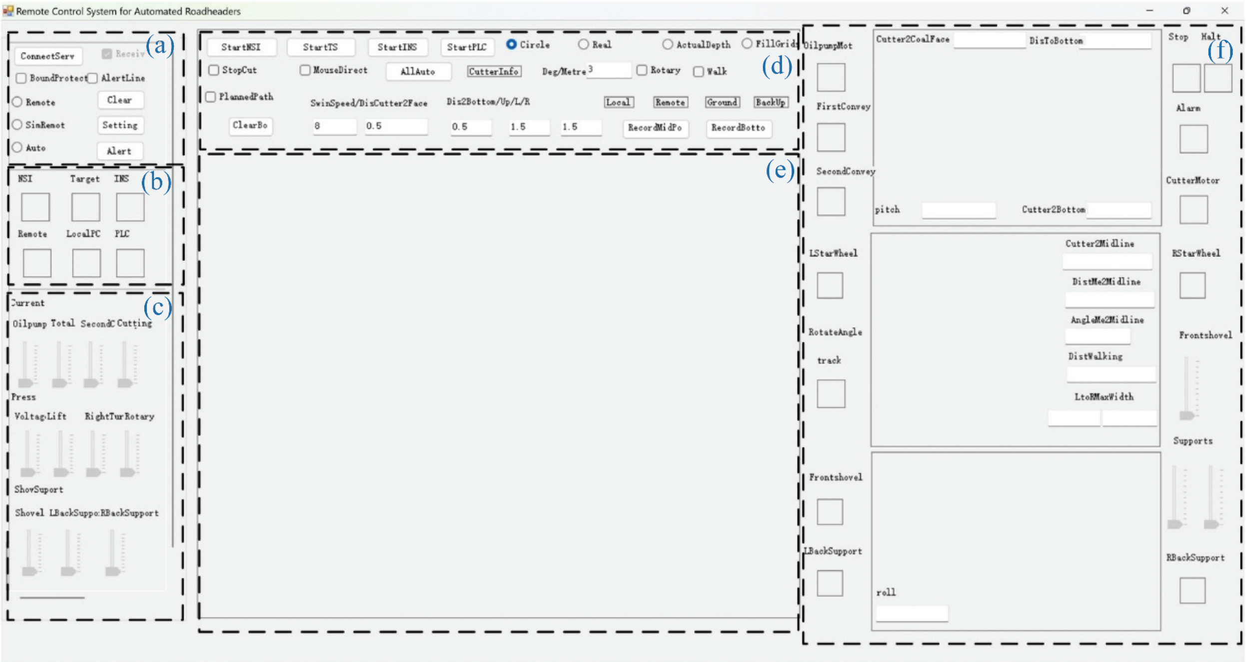

Figure 8. Interface of the remote-control system for unmanned roadheader tunneling.

corrections, and proceeds with self-coordinated cutting. The lower section presents key information about the working face, based on visualization establishment. This includes the border and alert line for potential cutting outside the designated area, division into two stages for cutting, the cutter's trajectory, its shape, and the penetration depth into the surface (Figure 8E). This section allows users to clearly observe that whether the cutter is operating within the designated area. The third section, located on the right side of the interface, primarily displays the relative position status between the roadheader and the tunneling work face from three different views in the center (Figure 8F). It also shows the operational status of the key roadheader components on both the left and right sides of this section (Figure 8F). Additionally, detailed real-time information is provided, including the distance from the cutter to the working face, the left and right borders, the bottom, as well as pose information and other relevant data (Figure 8F). This detailed information equips workers with the necessary insights to understand the exact operational status, enabling them to perform appropriate actions.

3. Machine experiments

Comprehensive experiments were conducted at the Zhaizhen Mine, a subsidiary of Xinwen Mining Group Co., Ltd., located in Tai'an, China. The geological structure of this mining area is relatively stable, with few faults and folds, minimizing the influence of tectonic activity. This stability provides favorable conditions for consistent mining operations. Mechanical tunneling with roadheaders is commonly employed in this region due to their suitability for local geological conditions, where the coal seams and surrounding rocks are generally stable and moderately hard.

Both ground and underground experiments were carried out utilizing the EBZ160 roadheader (Figure 9). Ground experiments were conducted at a roadheader manufacturing facility, while the underground experiments took place at Zhaizhen Mine.

The simulated tunnel, shown in Figure 9, has a rectangular cross-section with dimensions of 5.2 m in width and 4.5 m in height. The roadheader itself measures 1.805 m in width, 2.052 m in height, and 3.525 m in length. For underground testing, a real-world tunnel with identical dimensions to the simulated tunnel was used.

The experiments were divided into four distinct phases. The first three focused on evaluating specific primary functions, while the final phase tested the entire tunneling system.

3.1. Positioning and autonomous navigation

Our first test aimed to validate the robustness of the positioning and autonomous navigation function. To achieve this, we placed both the TS and the compass used for measuring the TS's orientation inside a specially designed explosion-proof enclosure (Figure 10). This enclosure is equipped with a window to allow the laser signals to pass through. During underground real-world tunneling experiments, it is important to note that this enclosure was suspended from a beam typically used for transporting

Figure 9. Roadheader for simulated tunneling test on factory ground.

Figure 10. Top: explosion-proof box housing the total station and compass in ground experiment; bottom: installation in a real-world underground tunneling scenario.

heavy equipment and was progressively moved forward every 40 m.

To enable precise localization of the roadheader's position, the target ball was securely affixed to a specific point on the rear of the roadheader (Figure 11, left). This setup allowed for accurate measurement of the roadheader's relative position, facilitating the determination of its absolute location.

The positioning experiments were divided into two distinct phases: the roadheader positioning test and the cutter head positioning test. For precise global positioning, a second TS, referred to as the "test TS,"

Figure 11. Left: target ball attached to the roadheader for precise positioning; right: cover used to prevent the target ball from being tracked by the total station (TS).

was aligned strictly with the north direction. This test TS functioned as a global positioning system, enabling direct measurements of results. The TS used for the roadheader positioning was designated as the "positioning TS." To ensure consistency, we first used the positioning TS to establish the location of the test TS before starting the experiments, ensuring that both TSs operated within the same coordinate system. The data from these two TSs were then compared to assess the accuracy of the positioning.

In line with the methodology described in (Cheluszka 2015), we subjected the cutter head positioning system, which is critical for tunneling operations, to further testing. The results, summarized in Table 2, show minimal positioning errors, highlighted in bold.

As previously emphasized, addressing the potential loss of the target is crucial to prevent accidents and ensure safety. To evaluate the system's response in such a scenario, we deliberately obstructed the target ball from the laser (Figure 11, right), rendering it undetectable. In response, the system effectively initiated a search for the target ball by systematically scanning within an increasing angle range. If the target ball remained elusive after three consecutive scans, the scanning process would halt, awaiting further instruction. It is important to note that before the target ball search began, the cutter head movement was paused. This testing confirmed the system's capability for robust positioning, successfully handling situations where the target is temporarily lost.

Moreover, in terms of autonomous navigation, once a target position is set, the roadheader should autonomously steer itself to reach that point by executing a sequence of movements, including rotations and straight walks. In our tests, after defining a target point and initiating the start command, the roadheader impressively adjusted its forward direction to align with the target and then proceeded to move until reaching the specified point. These autonomous navigation actions were seamlessly executed, exactly as intended within the system. For more details, interested readers can refer to the supplementary video.

Table 2. Cutter head positioning errors (m), (x_position, y_position, z_position) and (x_survey, y_survey, z_survey) represent the positioning and survey results, respectively.

x_position	y_position	z_position	x_survey	y_survey	z_survey
3.0241	7.3207	1.5136	3.0276	7.2855	1.5422
2.9647	7.2847	1.5136	3.0078	7.3174	1.4726
2.9072	7.2486	1.5136	2.9345	7.2769	1.4952
2.8512	7.2124	1.5137	2.8221	7.1751	1.5004
2.7968	7.1762	1.5137	2.8164	7.1591	1.5244
2.7439	7.1399	1.5137	2.7712	7.1271	1.5615
2.6922	7.1035	1.5137	2.6963	7.0587	1.4719
2.6419	7.0670	1.5138	2.6219	7.0447	1.4787
2.5928	7.0304	1.5138	2.6219	7.0561	1.5388
2.5448	6.9937	1.5138	2.4996	7.0316	1.5188
2.4979	6.9569	1.5138	2.5168	6.9108	1.5487
2.4520	6.9200	1.5139	2.4885	6.9407	1.5110
2.4072	6.8830	1.5139	2.3695	6.8630	1.5290
2.3632	6.8458	1.5139	2.4049	6.8861	1.4815
2.3202	6.8086	1.5140	2.2895	6.8076	1.5004
2.2781	6.7712	1.5140	2.2836	6.7567	1.4710
2.2368	6.7338	1.5141	2.2014	6.6993	1.5048
2.1964	6.6962	1.5141	2.1681	6.6484	1.5155
2.1567	6.6584	1.5141	2.1742	6.6678	1.4785
2.1178	6.6206	1.5142	2.1080	6.6245	1.4963
0.4697	2.8790	2.1265	0.4317	2.8590	2.1190
0.7671	2.9534	5.3653	0.7979	2.9240	5.4153
0.7618	3.0996	5.3650	0.7735	3.0721	5.3474
			0.02851	**0.02946**	**0.02808**

3.2. Visualization of the working face status

The visualization of the working face status during tunneling is shown in Figure 13A. At the center of the visualization, a gridded rectangular region precisely represents the real shape of the working face, whose exact width and height are shown. Additionally, the precise position of the cutter head on the working surface is shown. As previously elucidated, a gradual spectrum of gray colors, ranging from 0 to 255, mirrors the depth, moving from the minimum to the maximum depth level. These gray shades visually show the cutter head's penetration into the working surface (Figure 12C). Additionally, a palette of blue shades signifies the distance separating the cutter head from the working surface, transitioning from light blue to dark blue to indicate the degree of separation. The visual landscape includes distinctive red dotted rectangles marking boundaries that the cutter head must not cross (Figure 12A). Additionally, a dotted red line traces the central trajectory of the cutter head's movement, while a dotted blue curve indicates the direction of the cutter head's motion. A horizontal line divides the working face into upper and lower halves, facilitating a certain tunneling need.

Additionally, the interface displays key parameters that provide operators with quantifiable insights into the working status (Figure 12A). These parameters are accurately displayed, including the roll, yaw, and pitch angles of the roadheader, the distance traveled, and the horizontal distances from the cutter head center to key reference points, such as the front border, bottom, central axis, left border, and right border. Additionally, the horizontal distances between the boom and the left and right borders are provided. The display of key parameters is especially valuable for operators who might be unfamiliar with

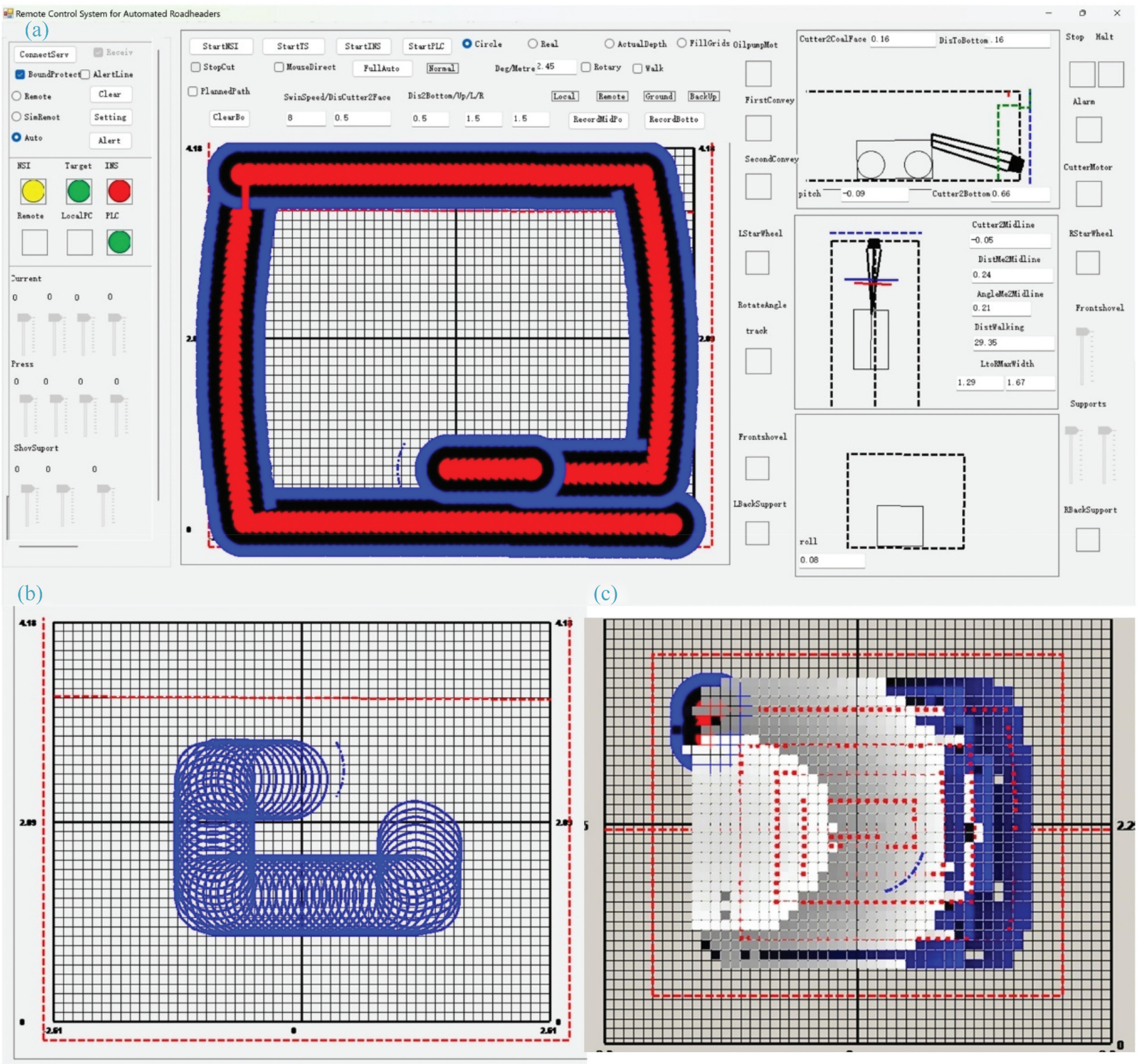

Figure 12. Visualization of working status for tunneling (A). B and C present different cutter head models and trajectories, respectively.

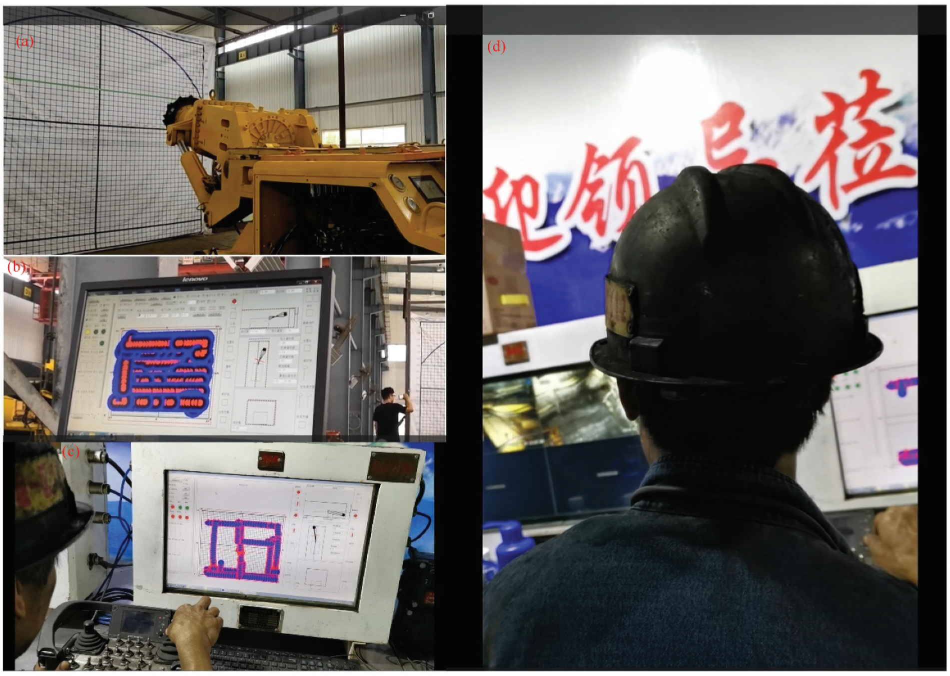

Figure 13. System operation during tunneling tests on ground and underground. A and B display the roadheader status and the system interface on the ground. C and D show workers operating the roadheader for underground unmanned tunneling from a location away from the working face. Different cutter trajectories can be observed in B and C. In D, the worker verifies if the system's visualization accurately reflects the real-world scenario captured by the cameras.

unmanned operation modes, enabling them to navigate roadheaders in a straightforward and understandable way.

3.3. Trajectory planning

In the trajectory planning tests, we confirmed that the generated trajectories accurately covered the entire working face area, regardless of its shape. The S-shaped, spiral, and free form trajectories are shown in Figure 13B, Figure 12A,C and Figure 13C, respectively. The system effectively followed these trajectories, ensuring complete coverage of the working face. These successful tests further validated the reliability and accuracy of the unmanned tunneling system's trajectory planning capabilities.

3.4. System application for tunneling

To comprehensively evaluate the whole system's performance, a tunneling experiment was conducted, covering the full process as defined by the system. Both manual and automatic operation modes were tested.

Initially, tests were conducted using remote controller-based tunneling, as shown in Figure 13. The results confirmed that tunneling with the remote controller is performed as expected. Operators were able to guide the roadheader's movements with ease and precision, resulting in the accurate formation of the tunnel's working face. Additionally, we tested a simulated remote controller, which demonstrated equal efficiency. It issued instructions at a predetermined frequency, enabling stable roadheader operations. When specific buttons were pressed, the instructions were accurately issued and executed as intended. These successful tests indicated that fully unmanned tunneling, without manual intervention, was feasible, with instructions being correctly issued and leading to autonomous completion of the tunneling task. Furthermore, the automatic operation-based tunneling is shown in Figures 13B and 12A,C, with the execution process shown in supplement video. According to statistical data and workers' feedback (Table 3), when the target ball was lost and could not be retracked, the system automatically halted, preventing any safety incidents. Throughout the tunneling process, the system operated in a stable and efficient manner, successfully completing the

Table 3. Performance evaluation of the system application for tunneling.

Operation mode	Deviation (<5cm)	Target ball loss reacquisition rate	Safety Incidents	Response time	System stability
Manual	Yes	>95%	0	Satisfied	Excellent
automatic	Yes	>95%	0	Satisfied	Excellent

tunneling task. This demonstrates the system's capability to perform tunneling tasks efficiently and correctly as intended whether controlled manually or automatically. However, while the system performed well, further evaluation and improvement of the response time and the accuracy are recommended.

4. Discussion

The developed system marks a significant breakthrough in overcoming the challenges of unmanned tunneling in China. Our comprehensive testing process has successfully validated several key aspects of the system.

4.1. The robust positioning method

Our method requires only a single point for positioning, in contrast to the method proposed by Fu et al. (2018), which necessitates three points, making its deployment more complex. Furthermore, our system effectively addresses the issue of target loss, ensuring robust positioning and seamless autonomous navigations. The target ball, which may suffer from loss due to dense dust, poses a significant challenge that can impede safe tunneling operations. Our method mitigates this issue by employing an optimized search algorithm to reacquire the target ball. If the target ball can't be tracked after three repeated search attempts, the cutting process is halted, and the system automatically stops, ensuring safe tunneling operations. Most existing studies, such as those by Deshmukh et al. (2020), do not address this issue, which suggests that our solution is more robust under these complex conditions.

4.2. The effectiveness of the visualization method

Precise modeling of the real-time cutter head working status in the working face enables highly accurate visualization of tunneling progress. In addition to the visual images, which accurately reflect the cutter head status in the working face and the relative position between the roadheader and tunnel, the system also provides the exact values necessary to define the working status. This comprehensive data set enhances operators' confidence in performing accurate tunneling operations. In contrast to the method by Liang and Lu (2010), which provide limited information, including a restricted set of values for describing tunneling status and visualization, our method provides more detailed data, allowing operators to place greater trust in the system and, consequently, improving operational efficiency.

4.3. Multiple trajectory planning and flexible operation manners

Our method provides multiple trajectory methods for both roadheaders and cutter head movement. For the roadheader, the planned trajectory enables automatic tunneling for tunnels of arbitrarily complexity, with a central axis line that can consist of multiple line segments. Regarding cutter head motion, our system supports more flexible movement options, including a mouse-click defined free-form route, S shape, spiral manner, and a manually cutting route recording-based method. Compared to the relevant methods (Liang and Lu 2010; Shen et al. 2014), our method is more practical and flexible. Additionally, the system supports various operation modes, including remote controller-based control, automatic cutter head motion for cutting, and fully automatic tunneling, which involves roadheader pose adjustment and movement for cutter head operation. Through tunneling experiments, the system has demonstrated its capability to achieve unmanned or remotely operated tunneling with both security and precision. To the best of our knowledge, no prior research has reported full automation or support for all these operation modes for roadheaders.

Despite these achievements, there are three points that could benefit from further improvement. First, automating the movement of the positioning box through the development of a robotic system would enhance overall efficiency. Second, the response time and the accuracy could be further evaluated and improved. Finally, exploring the integration of virtual reality-based 3D visualization presents a promising avenue for enhancing the autonomous operation of roadheaders.

5. Conclusions

In conclusion, our robust framework and sophisticated software have realized unmanned tunneling with roadheaders in China. By combining real-time positioning, multisensor data, high-precision modeling, and real-time visualization, we have achieved safe and precise unmanned tunneling. Both ground and underground experiments confirm its efficiency and effectiveness. The key achievements of our work are summarized as follows.

First, we developed a robust surveying method that combines a total station (TS) with a compass to measure its pose, enabling precise positioning using only a single point, with a solution to handle target loss.

Second, we developed a program-based remote controller that can coexist with the physical remote controller, offering flexibility and adaptability to multiple usage scenarios in complex environments.

Third, we designed a user-friendly visualization interface that provides sufficient information for cutting operations, significantly facilitating workers' tasks.

Finally, we implemented comprehensive safety and efficiency measures to ensure safe tunneling operations, providing reliable assistance to workers.

With these achievements, our framework holds great potential for automating unmanned tunneling with other underground machines, offering significant benefits across various industries. Its adaptability and effectiveness are valuable assets for advancing the field of unmanned tunneling.

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Notes on contributors

Changqing Yan received a PhD degree in cartography and geographic information systems from the Institute of Geographic Sciences and Natural Resources Research, Chinese Academy of Science, Beijing, China, in 2012. He is an associate professor in the College of Intelligent Equipment, Shandong University of Science and Technology. His research interests include intelligent equipment, geo-spatial data analysis, and smart agriculture.

Gang Zhao received a PhD degree in ecology from the Institute of Geographic Sciences and Natural Resources Research, Chinese Academy of Science, Beijing, China, in 2013. He is a professor in the College of Natural Resources and Environment, Northwest A&F University. His research interests include intelligent equipment, smart agriculture, and crop models.

Shengzhe Hou received a Bachelor degree of computer science from Shandong University of Science and Technology, Qingdao, China, in 2021. He is pursuing a PhD degree with the College of Computer Science and Engineering, Shandong University of Science and Technology. Her research interests include intelligent equipment, and 3D visualization.

Xinming Lu received a PhD degree in operations research and cybernetics from the Institute of Applied Mathematics, Chinese Academy of Science, Beijing, China, in 1994. He is a professor in the College of Computer Science and Engineering, Shandong University of Science and Technology. His research interests include intelligent equipment, smart mines, and geographic information systems.

ORCID

Changqing Yan http://orcid.org/0000-0002-0571-8025
Gang Zhao http://orcid.org/0009-0003-0619-5849
Shengzhe Hou http://orcid.org/0000-0003-1477-3074

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